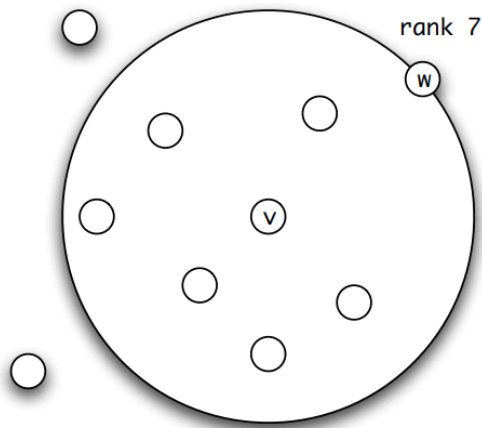
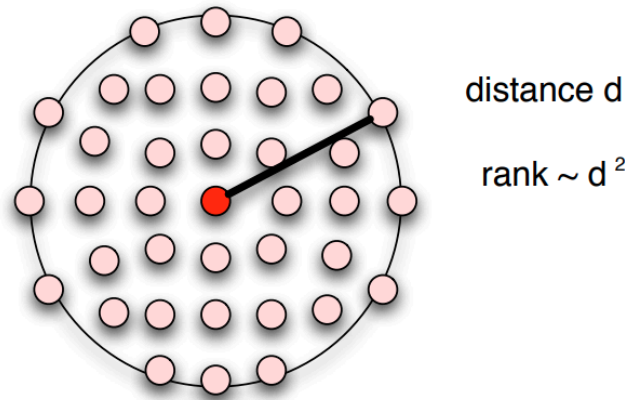


L11: Decentralized Network Search in Practice

Image Source: Textbook *"Networks, Crowds and Markets"* (<https://www.cs.cornell.edu/home/kleinber/networks-book>) by Easley and Kleinberg. Figure 20.9 and 20.10



(a) w is the 7th closest node to v .



(b) Rank-based friendship with uniform population density.

In the last few years, researchers have also investigated the efficiency of decentralized search empirically, mostly doing experiments in online social networks. There are a number of practical issues that have to be addressed however when we move from Kleinberg's highly abstract model to the real world.

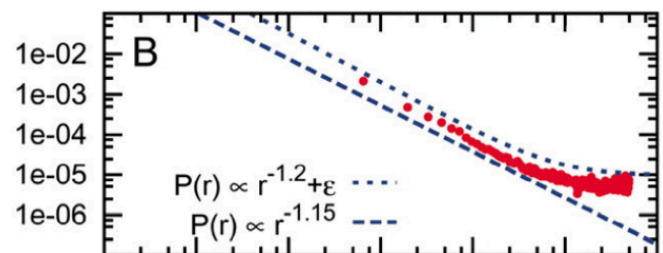
First, in practice, people do not "live" on a regular grid. In the US, for instance, the population density is extremely nonuniform, and so it would be unreasonable to expect that the $q=k=2$ result would also hold true about real social networks. One way to address this issue is to work with **"rank-based links"** rather than **"distance-based links"**. The rank of a node w , relative to a given node v , is the number of other nodes that are closer to v than w -- see Figure (a).

We can now modify Kleinberg's model so that the random links are formed based on node ranks, rather than rank distances. In Figure (b), we see that when nodes have a uniform population density, a node w at distance d from v will have a rank that is proportional to d^2 , since all nodes inside a circle of radius d will be closer to v than w .

In fact, it can be proven that, *for any population density* -- not just uniform -- if the random links are formed with a probability that is proportional to $\frac{1}{\text{rank}_v(w)}$, the resulting network will have minimum expected delivery time -- note that the exponent, in this case, is 1, not 2.

This result has been empirically validated with social experiments on an earlier social networking application called *LiveJournal* -- but similar experiments have been also conducted on Facebook data. Figure-c shows the probability of a friendship as a function of geographic rank on the blogging site LiveJournal. Note that the decay is almost inversely proportional to the rank.

Another way to generalize decentralized network search is by considering that people can belong to multiple different groups (*family, work, hobbies, etc*), and so the distance (*or rank*) between two network nodes can be defined *based on the group at which the two nodes are closest*. This group is referred to as the **"smallest focus"** that contains both nodes. This generalized distance metric has allowed researchers to examine more



(c) Rank-based friendship on LiveJournal

realistically whether real social networks have the optimal “**shortcut links**” to minimize the delivery time of the decentralized search, in terms of the corresponding exponent.

An interesting question, that is still largely an open research problem, is: *why is it that real social networks, both offline and online, have such optimal “**shortcut links**”?* Clearly, when people decide who to be friends with on Facebook or in real life, they do not think in terms of distances, ranks, and exponents!

One plausible explanation is that the network is not static – but instead, it evolves over time, constantly rewiring itself in a distributed manner, so that searches become more efficient. For instance, if you find at some point that you want some information about a remote country (e.g., *Greece*), you may try to find someone that you know from that country (e.g., *the instructor of this course*) and request that you get connected on Facebook. That link can make your future searches about that country much more efficient.